Zihe Zhou

123090915@link.cuhk.edu.cn https://zihezhou62.github.io/
Shenzhen, China

Education

The Chinese University of Hong Kong, Shenzhen Shenzhen, China
B.Sc. in Applied Mathematics Sep 2023–Jun 2027

- **GPA:** 3.803/4.0 **Major GPA:** 3.822/4.0 **Major Rank:** 7/88
- **Core Coursework:** Coding Theory, Optimization, Abstract Algebra, Real Analysis (Measure Theory), Probability and Statistics, Complex Variables, Ordinary Differential Equations
- Academic Performance Scholarship
- Dean's List (2023–2026)
- Honorable Mention, 2026 Mathematical Contest in Modeling, COMAP

University of Oxford Oxford, UK
Visiting Student Programme, St Edmund Hall Oct 2025–Jun 2026

- **Selected Coursework:** Information Theory, Quantum Information Theory, Quantum Theory, Graph Theory, Applied Partial Differential Equations, Stochastic Partial Differential Equations, and Number Theory.

Stanford University Palo Alto, USA
Summer Session: Programming Abstractions; Machine Learning Jun 2025–Aug 2025

Publications

Decoding Tanner Codes on Bipartite Two-Sided Vertex Expanders.
Zihe Zhou[*], Ningzi Wu[*], and Zitan Chen.
Manuscript under review at ITCS 2027.

New Generalization Bounds for Quantum Learning via f-Divergences and Maximal Leakage.
Zihe Zhou, Chun Hei Michael Shiu, Yanxiao Liu
Manuscript under review at ITW 2027

Research Experience

Coding Theory and Randomized Algorithms Sep 2024–Aug 2026
Supervised Research and Independent Study, advised by Prof. Zitan Chen, The Chinese University of Hong Kong, Shenzhen

Project 1: Tanner-Code Decoding on Bipartite Expanders Mar 2026–Aug 2026

- Developed a deterministic $O(n)$-time decoder correcting $\Omega(n)$ adversarial errors in Tanner codes defined on two-sided expanders.
- Extended the error-to-erasure approach to general binary inner codes using decoded-word disagreement and inner-code support completion.
- Proved that, under the additional right-expansion condition $\delta_R > 1 - 2/d$, the required left-expansion threshold can be reduced to $\delta_L > 1/(d_0 - 1)$, improving on the previous product-of-two sufficient condition for general inner codes.

Independent Study 2: Polar Codes and Information-Theoretic Foundations Feb 2025–May 2025

- Conducted a literature-based study of channel polarization, successive-cancellation decoding, and asymptotic reliability guarantees; presented key proofs and open problems.

Independent Study 3: Randomized Algorithms and Probabilistic Methods Sep 2024–Dec 2024

- Studied probabilistic analyses of randomized algorithms, including randomized quicksort and Karger's minimum-cut algorithm; completed formal proofs, presentations, and a summary report.

Generalization Bounds for Quantum Learning Jul 2026–Present
Supervised Research, advised by Dr. Yanxiao Liu, Imperial College London

- Replaced the Hölder relaxation in the final classical Rényi change-of-measure step of Warsi, Dasgupta, and Hayashi with exact binary data processing, proving a strictly tighter high-probability bound for every Rényi order $\beta > 1$ and confidence level $\delta \in (0, 1)$.
- Derived complementary bounds based on Sibson Rényi information, classical f-divergences, and maximal leakage.

Belief Propagation for Tensor Networks May 2026–Present
Supervised Research, advised by Prof. Shenghao Yang The Chinese University of Hong Kong, Shenzhen

- Developing network-dependent convergence conditions for belief propagation beyond strong injectivity, including singular positive-semidefinite messages and loopy geometries.
- Deriving error bounds for loop-corrected environments and repeated BP–gate–simple-update dynamics.

Teaching Experience

University Teaching Assistant, Calculus Jan–May 2025

- Led weekly tutorials,Designed and administered quizzes.

University Teaching Helper Sep–Dec 2024;Feb–May 2025

- Provided one-on-one and small-group support in undergraduate mathematics through the Engineering Faculty Help Desk.